Plot[InterpolatingFunction[Domain: {{0., 5.66 × 10^3}} Output: scalar][t],

{PBH_Trajectory_Plot t, 0, 5664.25 PBH_Trajectory_Plot},

{Frame → True, Axes → False, PlotStyle → {{ , AbsoluteThickness[1.5]}},

FrameStyle → Directive[, AbsoluteThickness[1]], FrameTicks →

{{Automatic, Automatic}, {Table[{i, i}, {i, 0, Ceiling[5664.25 PBH_Trajectory_Plot], 2}] ∪

$$\left\{\left\{\frac{1}{2}\left(\frac{24\,055\,290\,193}{5\,000\,000\,000}+\frac{3\sqrt{61\,517\,442\,918\,826\,886\,361}}{5\,000\,000\,000}\right),\ \bar{r}_0\right\}\right\},\ \text{Automatic}\}\},\ \text{FrameTicksStyle} \rightarrow$$

$$\text{Directive}[\ ,\ 11],\ \text{GridLines} \rightarrow \left\{\left\{\frac{1}{2}\left(\frac{24\,055\,290\,193}{5\,000\,000\,000}+\frac{3\sqrt{61\,517\,442\,918\,826\,886\,361}}{5\,000\,000\,000}\right)\right\},\ \text{Automatic}\right\},$$

GridLinesStyle → Directive[, Dashing[{Small, Small}], AbsoluteThickness[0.8]],

$$\text{FrameLabel} \rightarrow \{\{Numerical\bar{r}_{in}\},\ t\},\ \text{ImageSize} \rightarrow 550,\ \text{AspectRatio} \rightarrow \frac{1}{\text{GoldenRatio}},$$

PlotRange → All, PlotRangePadding → Scaled[0.04], Method → {DefaultBoundaryStyle → None},

Exclusions → None, OptionValue[PlotRangePadding], Prolog → {Opacity[0.15], ,

$$\text{Rectangle}\left[\{0,\ -10\,000\,000\,000\},\ \left\{\frac{1}{2}\left(\frac{24\,055\,290\,193}{5\,000\,000\,000}+\frac{3\sqrt{61\,517\,442\,918\,826\,886\,361}}{5\,000\,000\,000}\right),\ 10\,000\,000\,000\right\}\right]\}\}]$$